

Brian Mwai

Forward Deployed and Design Engineer

Independent practice, remote. I also teach AI, to beginners and to developers.

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I help companies turn their messy, scattered data into an AI system that answers the questions they actually make decisions on.

THE PRACTICE

Embed	Work inside the client's operation until the real process is visible, rather than the one written down.
Model	Turn what is there into an ontology and a knowledge graph: the entities the business actually runs on, joined, with every value recording where it came from.
Deploy	Put agents on top of that model, so answers are grounded in defined records rather than guessed from text.
Decide	Land on a decision the client acts on. The engagement is sold as that outcome, not as hours.

The modelling is the edge. Ontology and knowledge graph work is unusual for a solo engineer, and it is the layer Palantir sells as its Ontology product.

African mobility and logistics are evidence of what the method does, not the boundary of who it is for.

SKILLS

Ontology and graphs	Modelling a business as entities, relationships and events; knowledge graphs; recording where each figure came from so an answer can be traced back to a source.
AI, building	Assistants and agents that carry out multi-step work; question answering over a company's own documents (RAG, embeddings, vector search); extracting data from invoices, contracts and forms; voice interfaces; connecting models to existing systems (MCP, function calling); several models used together, with fallbacks. Claude, GPT, Gemini and open models.
AI, safety and trust	Checking model output against the modelled rules before anyone acts on it; a person approves anything consequential; measuring quality before and after release.
AI, with people	Training technical and non-technical teams; reviewing how a team works to find the steps worth automating; advising which tasks to give a model and which to leave alone.
Web and desktop	React, Next.js, TypeScript, Tailwind CSS, Node.js; desktop apps with Tauri 2 and Electron
Data	Convex, PostgreSQL, Python, Rust, mapping with OpenLayers, MapLibre and Mapbox
Design	Interface design, brand identity, design systems, Figma

TOOLS I WORK WITH

AI	Claude Code · Claude, GPT and Gemini APIs · MCP servers · Ollama for local models · ElevenLabs and fish.audio for voice · vector databases
Knowledge	Ontologies · knowledge graphs · Obsidian
Build and run	Git and GitHub · Zed · Vercel · Cloudflare · AWS · Convex · PostgreSQL
Design	Figma · Blender

WHAT I BUILD FOR CLIENTS

An assistant inside a tool your team already uses, answering questions from your own data · An agent that completes a repeated multi-step task, with a person approving what matters · A pipeline that reads documents into structured records · A data model that makes answers traceable · Training so your staff can run these tools without me

EXPERIENCE

Forward Deployed and Design Engineer

Ongoing

Independent practice, remote

- Embed with a client, model their domain as an ontology and knowledge graph, deploy agents on top of it, and land on a decision they act on. Sold as the deployed outcome rather than as hours.
- Run AI training alongside the engagements, for technical and non-technical staff.
- Lead engagement: Odyssey, delivered for Celeri Group.

Technical Lead and Co-founder

Jun 2026 – Present

Revlog, Nairobi · revlog.africa

- Battery collection and recycling business, seven co-founders. I own the technical side: data model, web portal and the tools field staff use.
- Built CycleTrack, the custody record that follows each battery from producer to offtaker, with consoles for producers, offtakers, regulators, field staff and admin on one Convex backend. cycletrack.revlog.africa
- Built the public collection portal, where anyone can book a collection, report an unsafe battery or find a drop-off point. portal.revlog.africa
- Pitched the company in the final of the ACEP Innovation Challenge at the Future of Energy Conference, Accra, August 2026.

Software Engineer, contract

Mar 2026 – Jul 2026

Celeri Group, Nairobi

- Sole developer on Odyssey, a map-based tool for investors assessing transport routes in Kenya. Delivered a working demo and investor pack in about two months.
- Built the map covering all 47 counties: routes, roads, operators, schools, hospitals and fuel stations, up to 1,789 stations shown at once.
- Designed the data model, 22 record types, with every value recording where it came from and when, so figures can be traced to a source.

Co-founder and Design Engineer

Jul 2025 – Present

AXI Mobility, Nairobi and Delaware · aximobility.com

- Fleet management software for African transport operators. I do the design and the engineering: brand, design system, operator dashboard, driver app and website.
- Built an AI assistant into the dashboard that answers questions about the fleet and proposes actions for a person to approve. Set up vehicle tracking that works with hardware customers already own, designed for 10,000 vehicles.
- Represented the company at the Africa Climate Summit 2025 in Ethiopia, in the African Union Young Innovation Hub.

Software Engineer

May 2025 – Aug 2025

Kenya Space Agency, Nairobi

- Built AIMS, a satellite-based crop monitoring tool for Murang'a County, using React, TypeScript and OpenLayers. Land-cover comparison across 2022 and 2024, crop classification, weather and market prices by subcounty. Works offline, in English and Swahili.

Founder and Designer

2023 – Present

Formartly, Nairobi

- Creative studio. Brand identity, logo and merchandise design for clients, including merchandise for Sauti Sol.

EDUCATION

The Catholic University of Eastern Africa, Nairobi

2022 – expected 2026

BSc Physics, final year. Supervisor: Dr. Songa Mutambi.

- Dissertation: *Measuring the External Field: A Cortical-Proximal Content Observable and the Mean-Field Bound on Media-Driven Opinion Change*.
- Built the instrument it rests on. Meta's TRIBE v2 model predicts how the cortex responds to a piece of media, and the index is the difference between networks linked to emotion and to reasoning. Scanned a 400-item media corpus; a second full scan agrees at ICC 0.87. From the measured spread, derived the minimum strength any content signal needs to tip a group's opinion in a mean-field model.

Lead Designer, Google Developer Student Clubs, 2024 to January 2025.

SELECTED PROJECTS

Cascade Bridge · naturecipher-drought.pages.dev

Drought forecasting for Kenyan counties up to three months ahead, from seasonal weather forecasts rather than satellite images. Top 10, IGAD ICPAC hackathon, 2026, with Nature Cipher.

Monarch · monarch-4iy.pages.dev/corpus · [source](#)

The instrument behind the dissertation, live with its measured corpus. It rates how far a piece of written media leans on emotion rather than reasoning, from Meta's TRIBE v2 prediction of the cortical response. The site shows all 400 scanned items with their scores and the predicted response on the cortical surface. It rates content, never a person. Open source.

Rovify · rovify.io

Events platform on Base. Third place globally, USD 20,000, Base Batches 001, 2025, from over 1,200 entries. Winner in the Africa round, with two teammates.

Oryn · oryn.cc

Reads invoices and contracts and turns them into payment links. Third place, AI Mashinani, run by the Claude community in Kenya, 2026.

PayStream

Payroll paid in stablecoin, with the compliance and conversion steps automated. First place, Qdrant track, AI Genesis, Dubai, 2025.

Tollgate

Lets a website charge AI bots per request in USDC instead of blocking them, using HTTP 402 payments on Circle's Arc network, with a publisher dashboard and middleware for Express and Hono. Built for Circle's Agentic Economy on Arc hackathon, April 2026.

TOOLS I HAVE BUILT

FDE Accelerator

The engine behind the practice. Turns a client's scattered data into an ontology and knowledge graph, sharpened on each paid engagement. Rust backend, running as the FDE Workbench inside Wispy Console. Kept internal rather than sold as a product.

Wispy Console

Windows workspace for building with coding agents: terminals, projects, per-project secrets encrypted by Windows, a code graph, and a task board that Claude Code, Codex, Gemini and OpenCode write to. Rust and React on Tauri 2, shipped as an installer at version 1.1.0. Closed source.

Wispy · wispy.cc · npm · source

Autonomous AI agent platform. One command gives you a running agent that can act and communicate across 11 channels. Built for the Google Gemini hackathon.

Wispy Desktop

Meeting assistant for Windows. Transcribes both sides of a call on the device with Whisper and answers questions with Claude from your own notes, searched with embeddings that also run locally. Electron and React.

codegraph · github.com/brn-mwai/codegraph

Turns a codebase into a map an AI can navigate. Parses the repo with tree-sitter into a typed graph, so an agent can find where something is defined and what breaks if it changes, instead of searching blindly.

Scribe · github.com/brn-mwai/scribe

Keeps a repository's agent instructions accurate on its own. When the code changes and the docs go stale, it notices and opens a merge request to fix them.

Agent Reach

Gives an AI agent access to the internet across 15 platforms through one interface, with fallbacks when a source is unavailable.

RECOGNITION AND PROGRAMMES

AXI Mobility	KES 300,000 proof-of-concept grant, Kenya Climate Innovation Center, 2026 · Timbuktoo Green Tech Hub · NVIDIA Inception · 500 Global Pre-Acceleration Academy, Nairobi, October 2025 · African Union Young Innovation Hub, Africa Climate Summit 2025, Ethiopia
Revlog	Finalist, ACEP Innovation Challenge, Future of Energy Conference, Accra, August 2026 · Incubation and grant support from LOT29, GIZ, Siemens Stiftung and Seedstars

LANGUAGES AND INTERESTS

Languages	English (native) · Kiswahili (fluent)
Interests	Drawing and illustration, national top 10 in Red Bull Doodle Art Kenya, 2023, from over 8,000 entries. Guitar. Hiking and the outdoors. Film.

REFEREES

Kenneth Kanda Deputy Director, Digital Infrastructure and Services Kenya Space Agency kenneth.kanda@ksa.go.ke +254 710 300235	Dr. Songa Mutambi Dissertation supervisor, Department of Natural Sciences The Catholic University of Eastern Africa cmutambi@cuea.edu +254 724 508575
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